

Unified Mechanics

*A unified framework for physical state, reflexive structure,
field law, and empirical programme*

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ABSTRACT

Unified Mechanics is presented here as a single dependency-ordered theory. It uses one notation and one claim vocabulary, and it separates mathematical necessity, physical principles, realisation bridges, and empirical tests so that each result carries exactly the strength established for it.

The theory is organised by a three-stage law. First, a physical state is fixed by the futures it must determine: carriers are physically identical for a declared task exactly when every admissible continuation and readout gives the same law. The resulting quotient is the unique coarsest lossless state. Second, the declared continuations descend uniquely to that quotient, so task-relative dynamics is not an additional ornament but the action already carried by the state. Third, when that dynamically sufficient state must be realised as a finite persistent record, the realised history minimises cumulative dynamically necessary boundary.

The self-description sector supplies a further exact closure. A reflexive memory-one task that preserves present resolution and retained context requires exactly two channels. Under minimality and exact one-copy semantics the update is uniquely $(R, C) \mapsto (R + C, R)$. Its positive projective fixed ray is φ , its one-generator continuation monoid is free, and a two-sided physical reading fixes $r = (2\varphi)^{-1}$. The exact partition $1 = (1 - r)^2 + 2r(1 - r) + r^2$ then yields the propagation, interface, and retention weights identified in the physical theory as Light, Boundary, and Matter.

Unified Mechanics treats the foundational tautology as a source of power rather than an embarrassment. A tautology can be empty when it merely renames an assumption. Here it is constraining because its universal property excludes every state compression, dynamical reduction, action representative, and physical readout that erases a future-active distinction. The mathematical necessity, the physical principles, the carrier identifications, and the empirical tests are separated so none can borrow strength from another and none is needlessly weakened.

Contents

1	THE CONTROLLING STATEMENT	4
2	WHY THE TAUTOLOGY IS SUBSTANTIVE	5
3	STRUCTURE OF THE THEORY	6
4	ONE VOCABULARY	6
5	CLAIM STRENGTH WITHOUT CLAIM INFLATION	7
6	THE MASTER LAW IN COMPACT FORM	8
7	HOW THE THEORY IS ORGANISED	8
8	THE THEORY'S STRONGEST PRESENT CLAIMS	9
9	READING RULE FOR ALL APPLICATIONS	9
10	SCOPE AND SCIENTIFIC STATUS	10

1 The controlling statement

MASTER STATEMENT

A realised physical system is represented by the coarsest retained distinction sufficient to determine every declared admissible future readout. The declared continuations act uniquely on that state. Among finite persistent realisations that preserve the same future-active content, the realised history is the one with least cumulative dynamically necessary boundary.

In the reflexive, minimal, exact-copy self-description sector, the unique generator is

$$F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

its positive projective fixed ray is φ , and two-sided realisation gives

$$r = \frac{1}{2\varphi}, \quad (W_L, W_B, W_M) = ((1-r)^2, 2r(1-r), r^2).$$

This statement is the core law of Unified Mechanics. Its sentences are not independent slogans. They form a dependency chain:

$$\begin{aligned} \text{future determination} &\implies \text{canonical state} \implies \text{induced dynamics} \\ &\implies \text{dynamical sufficiency} \implies \text{least-boundary realisation.} \end{aligned} \tag{1}$$

The reflexive closure is a distinguished sector of the same chain:

$$\begin{aligned} \text{self-recording task} &\implies \text{two-channel state} \implies \text{whole-shift generator} \\ &\implies \varphi \implies r \implies (W_L, W_B, W_M). \end{aligned} \tag{2}$$

2 Why the tautology is substantive

The first sentence of the theory is deliberately tautological in the precise sense that a physical state for a task is defined by what must remain sufficient for that task. Its power comes from the word *coarsest*. Once the admissible futures and readouts are declared, there is exactly one maximal identification compatible with all of them. Every valid model must refine it; every valid readout must factor through it; and every continuation that belongs to the declared task must descend to it.

RESULT

The foundational tautology does not predict which task Nature declares. It does determine, without approximation, the unique state space and induced dynamics of any declared task. That is a nontrivial universal constraint on representation.

This distinction prevents two opposite errors. The first is to dismiss the result as “only definitional” and miss the factorisation, uniqueness, and obstruction theorems. The second is to claim that a universal property alone selects a particular physical carrier. The theory gives the tautology its full force while keeping the physical selection problem visible.

3 Structure of the theory

TABLE 1. Controlling location of every major result.

Paper	Governing question	Controlling result
01	What is the physical state of a declared task?	Future-law equivalence produces the unique coarsest lossless quotient; the continuation monoid descends uniquely; retained observation exposes the reversible core.
02	Which update closes a minimal self-description?	Present resolution and retained context require two channels; exact copy semantics selects the whole-shift matrix, φ , r , and the three exact path weights.
03	Is a Lagrangian the physical law?	No. Dynamics fixes an equivalence class of permissible actions; complete equivalence also preserves constraints, zero modes, boundary data, and records.
04	How is an abstract sufficient state physically realised?	Persistent finite realisation minimises cumulative necessary interface; the static sector obeys a mean-curvature field equation and the covariant sector is a world-tube action.
05	How does one theory reach physical sectors and tests?	Every sector is compiled through one carrier-continuation-record dictionary; completed calculations are separated from open carrier laws and prospective falsification.

4 One vocabulary

Distinction.

A difference that changes at least one declared admissible future law.

Identity.

Equality in the canonical task quotient: two carriers are the same physical state when no declared future can separate them.

Relation.

A continuation, comparison, coupling, or readout through which distinctions affect later law.

Record.

A distinction that remains recoverable under the relevant future continuation.

Object.

A recoverable, forward-closed, recurrent organisation in the quotient state space.

Boundary.

The operational interface at which one task-equivalence class can be separated from another; in a finite physical realisation it is the carrier of conversion burden.

Observation.

A retained physical correlation whose alternatives remain discriminable after the

declared future.

Action.

A representative whose stationary histories realise the same physical dynamics and boundary ledger; the representative is not unique.

Light.

The pure propagation path in the two-rate realisation.

Boundary.

The mixed path, occurring in both orders, that mediates propagation and retention.

Matter.

The pure retention path in the two-rate realisation.

The same word *boundary* appears once operationally and once as a named physical channel. These are not competing meanings. The operational definition says what a boundary does: it carries a future-active separation. The channel identification says which exact path class realises that role in the reflexive two-rate sector.

5 Claim strength without claim inflation

Unified Mechanics uses six labels.

Label	Meaning
Exact theorem	A mathematical consequence of explicitly stated data.
Derived consequence	A corollary of earlier theorems in the theory.
Physical principle	A law asserted by Unified Mechanics and exposed to physical rejection.
Realisation bridge	An identification between an exact structure and a physical carrier, process, or coefficient.
Empirical result	A completed calculation or data comparison with a declared input ledger.
Open test	A prospective calculation or observation whose result is not fixed in advance by the theory or its calibration data.

This vocabulary is not a hierarchy from “strong” to “weak”. An exact theorem can be physically irrelevant if its premises are never realised. A physical principle can be scientifically powerful even though it is not a theorem, because it rules out worlds. A realisation bridge can be decisive if it survives independent measurement. The labels indicate where the risk sits.

6 The master law in compact form

Let $\mathcal{C}$ be the complete carrier space, let $\mathcal{T}$ be the declared family of continuation-readout tasks, and let

$$c \sim_{\mathcal{T}} c' \iff \mathcal{L}_{\tau}(c) = \mathcal{L}_{\tau}(c') \text{ for every } \tau \in \mathcal{T}.$$

The physical state is $\mathcal{Q}_{\mathcal{T}} = \mathcal{C}/\sim_{\mathcal{T}}$. If the continuations form a monoid $\mathbf{M}$ and $\mathcal{T}$ is closed under their action, then there is a unique induced action $\bar{\Phi}$ satisfying

$$q \circ \Phi_m = \bar{\Phi}_m \circ q.$$

A finite physical realisation is an admissible history $\Omega(t)$ that preserves this factorisation and every future-active boundary, order, phase, topology, gauge, and record datum. Unified Mechanics selects

$$\Omega_* = \arg \min_{\Omega \in \mathfrak{A}_{\mathcal{T}}} \int_{t_0}^{t_1} \int_{\partial^* \Omega(t)} \sigma_I(x, n, t) d\mathcal{H}^{d-1} dt. \quad (3)$$

For homogeneous isotropic realisation, $\sigma_I = \kappa$, the fixed-volume Euler equation is

$$\kappa H = \lambda_V. \quad (4)$$

In the exact Light-Boundary-Matter calibration,

$$\kappa = W_B \eta, \quad W_B = 2r(1 - r), \quad r = (2\varphi)^{-1}. \quad (5)$$

Equations (3)–(5) are the compact field law of Unified Mechanics. The quotient determines what may not be erased; the variational law determines how that retained state is realised.

7 How the theory is organised

The theory is organised by dependency rather than by topic or historical order. The following principles determine the placement and interpretation of its results.

1. The future-determination quotient is the foundation. Distinction, observation, objecthood, process, boundary, and compiler are not parallel axioms.
2. Induced dynamics is placed immediately after the quotient. It is the same universal construction carried through continuation closure.
3. Self-description is a physical sector with a named reflexivity premise, not a second universal foundation.
4. The golden ratio is derived exactly inside that sector. Its physical universality is a realisation claim, not hidden inside the algebra.
5. Equivalent action and least-boundary action are separated. The first classifies representations of a fixed dynamics; the second selects a physical realisation among dynamically sufficient histories.

6. Light, Boundary, and Matter are one physical dictionary for the exact pure-mixed-pure partition. Other three-way descriptions are translations, not additional triplets.
7. Application papers are not independent foundations. Each must identify its carrier, continuation, readout, action class, calibration, and rejection condition.

8 The theory's strongest present claims

The strongest claims are not the largest numerical lists. They are the claims with the longest reach and the least representational freedom.

1. **Canonical state.** Once a physical task is declared, the coarsest exact state is unique.
2. **Induced dynamics.** Once the task is continuation-closed, its state carries exactly one descended action.
3. **Reflexive closure.** Once self-description, minimality, memory one, and exact copy semantics are declared, the generator is unique and the golden fixed ray follows.
4. **Action non-uniqueness.** A physical dynamics does not privilege a unique written Lagrangian; equivalence must be tested at the level of histories, constraints, boundaries, and records.
5. **Least-boundary realisation.** Once finite persistent conversion is local, positive, capacity-limited, and task-equivalent, the realisation law is a weighted perimeter action with exact geometric consequences.

Unified Mechanics is therefore strong in a precise way. It does not say that one unexplained number generates every physical constant. It says that physical representation, continuation, and realisation are constrained in sequence, and that a distinguished reflexive sector closes to a unique arithmetic and geometry.

9 Reading rule for all applications

Every application in Paper 05 is read through the same eight questions:

1. What is the complete carrier?
2. Which futures and readouts define the task?
3. What quotient is the physical state?
4. Which continuation acts on it?
5. Which distinctions remain future-active?
6. Where is the physical interface and what density does it carry?
7. Which action representative and calibration are used?
8. Which measured record would reject the proposal?

A calculation that cannot answer these questions is not yet a Unified Mechanics prediction, even if it contains φ . A calculation that can answer them belongs to the same theory even when its carrier is very different from another application.

10 Scope and scientific status

RESULT

Unified Mechanics is presented here as a coherent theory statement: it has one state principle, one induced dynamics, one reflexive closure, one action-equivalence doctrine, one physical realisation law, and one empirical compiler. It remains scientifically open wherever a carrier law, coefficient map, or prospective observation has not yet been independently fixed.

The exact theorems establish the representation and continuation structure of declared tasks. The physical principles and realisation bridges state how that structure is proposed to operate in Nature. The empirical programme then tests those bridges without altering the foundational quotient to accommodate a result.